

Missing-verb illusion in Turkish center-embeddings? An investigation of case interference and phrase lengths

Özge Bakay & Nazik Dinçtopal Deniz

Boğaziçi University, Turkey

AMLaP 2020





Introduction



- This study investigates doubly center-embedded structures with relative clauses (2-CE-RCs) in Turkish.
 - *E.g., The rat that the cat that the dog chased ate died.* (Chomsky & Miller, 1963, p. 286)
- 2-CE-RCs are reported to be extremely difficult to process due to (among many other factors)
 - working memory limitations (Gibson, 1998; 2000)
 - **similarity-based interference** (Lewis & Vasishth, 2005)
 - **non-optimal prosodic phrase lengths** (Fodor, 2013)
- Due to their extreme processing difficulty, the omission of the second verb in 2-CE-RCs can be unnoticed, resulting in a grammaticality illusion:
 - *The patient who the nurse who the clinic had hired ~~admitted~~ met Jack.* (Frazier, 1985, p. 178)
- **Missing-verb illusion** has been confirmed in verb-initial languages;
 - English (e.g., Christiansen & MacDonald, 2009; Gibson & Thomas, 1999)
 - French (Gimenes et al., 2009)
- Findings on verb-final languages are inconclusive;
 - they either support its presence, e.g., German (Bader, 2015; Häussler & Bader, 2015)
 - or not, e.g., German and Dutch (Frank & Ernst, 2018; Frank et al., 2016; Vasishth et al., 2010)
- In verb-final languages more robust predictions for upcoming verbs may prevent the illusion.



The Present Study



- Turkish, a highly inflectional and head-final language, can provide further data on similarity-based interference, prosodic phrase lengths and missing-verb illusion.
- The present study investigates:
 - if the processing difficulty of Turkish center-embeddings can be facilitated by similarity-based interference and/or prosodic phrase lengths,
 - if there is a missing-verb illusion in Turkish center-embeddings,
 - if the existence of missing-verb illusion could be modulated by similarity-based interference and/or prosodic phrase lengths.
- Two acceptability judgment tasks were conducted:
 - Expt. 1: tests similarity-based interference and missing-verb illusion, 80 participants
 - Expt. 2: tests prosodic phrase lengths and missing-verb illusion, 38 participants



Expts. 1 & 2: Materials & Procedure



- Experimental sentences were Turkish 2-CE-RCs nested as a complement clause inside a matrix clause:

Ø	[Marangoz-lar-ın	[nakliyeciler-in	[kiracı-nın	gri	koltuğ-u	yerleştir-diğ-i	oda-ya]
Pro	carpenter-PL-GEN	mover-PL-GEN	tenant-GEN	gray	sofa-ACC	place-PART-3SG	room-DAT
	NPI	NP2	NP3			VP3	
taşı-dık-ları	dolab-ı]	kur-duk-ları-nı]					
move-PART-3PL	cabinet-ACC	set.up-VN-3PL-ACC					
VP2		VPI					

'I know that the carpenters set up the cabinet that the movers moved to the room that the tenant placed the gray sofa in.'

- Expt. 1:** four conditions manipulating
 - grammaticality (grammatical vs. ungrammatical due to a missing VP2 (in blue above))
 - syntactic interference (high vs. low interference manipulated with case marking on the subject NPs)
- Expt. 2:** four conditions manipulating
 - grammaticality as in Expt. 1
 - prosodic phrase lengths (encouraging vs. discouraging phrase lengths for the optimal phrasing of the sentence)
- Comprehension questions about different subject-verb relations followed the sentences.
- 24 experimental sentences & 36 fillers in each experiment
- Task: rate the sentences on a scale of 1 to 5 (5, acceptable)



Results



	Expt. 1			Expt. 2		
	LowInt	HighInt	Accuracy	ENC	DISC	Accuracy
Grammatical	2.59 (.05)	2.86 (.05)	78%	2.87 (.09)	2.69 (.09)	81%
Ungrammatical	2.40 (.08)	2.48 (.08)	65%	2.63 (.08)	2.54 (.08)	66%

Cumulative link mixed effects models with R package *ordinal* (Christensen, 2018) showed:

Expt. 1: HighInt > LowInt $(\beta = .40, SE = .13, z = 3.04)$

Grammatical > Ungrammatical $(\beta = .36, SE = .17, z = 2.15)$

HighInt: Grammatical > Ungrammatical $(\beta = .54, SE = .22, z = 2.45)$

LowInt: Grammatical = Ungrammatical $(\beta = .14, SE = .22, z = .67)$

Expt. 2: ENC > DISC $(\beta = .25, SE = .14, z = 1.76)$

Grammatical > Ungrammatical $(\beta = .36, SE = .19, z = 1.88)$

ENC: Grammatical > Ungrammatical $(\beta = .48, SE = .25, z = 1.92)$

DISC: Grammatical = Ungrammatical $(\beta = .21, SE = .20, z = 1.03)$



Results



	Expt. 1			Expt. 2		
	LowInt	HighInt	Accuracy	ENC	DISC	Accuracy
Grammatical	2.59 (.05)	2.86 (.05)	78%	2.87 (.09)	2.69 (.09)	81%
Ungrammatical	2.40 (.08)	2.48 (.08)	65%	2.63 (.08)	2.54 (.08)	66%

Cumulative link mixed effects models with R package *ordinal* (Christensen, 2018) showed:

Expt. 1: HighInt > LowInt $(\beta = .40, SE = .13, z = 3.04)$

Grammatical > Ungrammatical $(\beta = .36, SE = .17, z = 2.15)$

HighInt: Grammatical > Ungrammatical $(\beta = .54, SE = .22, z = 2.45)$

LowInt: Grammatical = Ungrammatical $(\beta = .14, SE = .22, z = .67)$

Expt. 2: ENC > DISC $(\beta = .25, SE = .14, z = 1.76)$

Grammatical > Ungrammatical $(\beta = .36, SE = .19, z = 1.88)$

ENC: Grammatical > Ungrammatical $(\beta = .48, SE = .25, z = 1.92)$

DISC: Grammatical = Ungrammatical $(\beta = .21, SE = .20, z = 1.03)$



Results



	Expt. 1			Expt. 2		
	LowInt	HighInt	Accuracy	ENC	DISC	Accuracy
Grammatical	2.59 (.05)	2.86 (.05)	78%	2.87 (.09)	2.69 (.09)	81%
Ungrammatical	2.40 (.08)	2.48 (.08)	65%	2.63 (.08)	2.54 (.08)	66%

Cumulative link mixed effects models with R package *ordinal* (Christensen, 2018) showed:

Expt. 1: HighInt > LowInt $(\beta = .40, SE = .13, z = 3.04)$

Grammatical > Ungrammatical $(\beta = .36, SE = .17, z = 2.15)$

HighInt: Grammatical > Ungrammatical $(\beta = .54, SE = .22, z = 2.45)$

LowInt: Grammatical = Ungrammatical $(\beta = .14, SE = .22, z = .67)$

Expt. 2: ENC > DISC $(\beta = .25, SE = .14, z = 1.76)$

Grammatical > Ungrammatical $(\beta = .36, SE = .19, z = 1.88)$

ENC: Grammatical > Ungrammatical $(\beta = .48, SE = .25, z = 1.92)$

DISC: Grammatical = Ungrammatical $(\beta = .21, SE = .20, z = 1.03)$



Results



	Expt. 1			Expt. 2		
	LowInt	HighInt	Accuracy	ENC	DISC	Accuracy
Grammatical	2.59 (.05)	2.86 (.05)	78%	2.87 (.09)	2.69 (.09)	81%
Ungrammatical	2.40 (.08)	2.48 (.08)	65%	2.63 (.08)	2.54 (.08)	66%

Cumulative link mixed effects models with R package *ordinal* (Christensen, 2018) showed:

Expt. 1: HighInt > LowInt $(\beta = .40, SE = .13, z = 3.04)$

Grammatical > Ungrammatical $(\beta = .36, SE = .17, z = 2.15)$

HighInt: Grammatical > Ungrammatical $(\beta = .54, SE = .22, z = 2.45)$

LowInt: Grammatical = Ungrammatical $(\beta = .14, SE = .22, z = .67)$

Expt. 2: ENC > DISC $(\beta = .25, SE = .14, z = 1.76)$

Grammatical > Ungrammatical $(\beta = .36, SE = .19, z = 1.88)$

ENC: Grammatical > Ungrammatical $(\beta = .48, SE = .25, z = 1.92)$

DISC: Grammatical = Ungrammatical $(\beta = .21, SE = .20, z = 1.03)$



Results



	Expt. 1			Expt. 2		
	LowInt	HighInt	Accuracy	ENC	DISC	Accuracy
Grammatical	2.59 (.05)	2.86 (.05)	78%	2.87 (.09)	2.69 (.09)	81%
Ungrammatical	2.40 (.08)	2.48 (.08)	65%	2.63 (.08)	2.54 (.08)	66%

Cumulative link mixed effects models with R package *ordinal* (Christensen, 2018) showed:

Expt. 1: HighInt > LowInt $(\beta = .40, SE = .13, z = 3.04)$

Grammatical > Ungrammatical $(\beta = .36, SE = .17, z = 2.15)$

HighInt: Grammatical > Ungrammatical $(\beta = .54, SE = .22, z = 2.45)$

LowInt: Grammatical = Ungrammatical $(\beta = .14, SE = .22, z = .67)$

Expt. 2: ENC > DISC $(\beta = .25, SE = .14, z = 1.76)$

Grammatical > Ungrammatical $(\beta = .36, SE = .19, z = 1.88)$

ENC: Grammatical > Ungrammatical $(\beta = .48, SE = .25, z = 1.92)$

DISC: Grammatical = Ungrammatical $(\beta = .21, SE = .20, z = 1.03)$



Results



	Expt. 1			Expt. 2		
	LowInt	HighInt	Accuracy	ENC	DISC	Accuracy
Grammatical	2.59 (.05)	2.86 (.05)	78%	2.87 (.09)	2.69 (.09)	81%
Ungrammatical	2.40 (.08)	2.48 (.08)	65%	2.63 (.08)	2.54 (.08)	66%

Cumulative link mixed effects models with R package *ordinal* (Christensen, 2018) showed:

Expt. 1: HighInt > LowInt $(\beta = .40, SE = .13, z = 3.04)$

Grammatical > Ungrammatical $(\beta = .36, SE = .17, z = 2.15)$

HighInt: Grammatical > Ungrammatical $(\beta = .54, SE = .22, z = 2.45)$

LowInt: Grammatical = Ungrammatical $(\beta = .14, SE = .22, z = .67)$

Expt. 2: ENC > DISC $(\beta = .25, SE = .14, z = 1.76)$

Grammatical > Ungrammatical $(\beta = .36, SE = .19, z = 1.88)$

ENC: Grammatical > Ungrammatical $(\beta = .48, SE = .25, z = 1.92)$

DISC: Grammatical = Ungrammatical $(\beta = .21, SE = .20, z = 1.03)$



Conclusions



- **Similarity-based interference:**

- Syntactic interference affected processing Turkish center-embeddings but in the unpredicted direction. Why?
 - judgment (i.e., offline) data may reveal limited effects of case interference (e.g., Avetisyan et al., 2020)
 - the non-factive reading in the LowInt conditions may have caused an extra processing difficulty (Özyıldız, 2017)
- An eye-tracking experiment with sentences excluding non-factive interpretation verbs showed the predicted pattern (data in the Q&A session).

- **Prosodic phrase lengths:**

- Optimal and relatively balanced phrase lengths eased the processing of Turkish center-embeddings (Fodor, 2013)

- **Missing-verb illusion:**

- Overall no missing-verb illusion in Turkish 2-CE-RCs; verb-final languages may be less susceptible to the illusion (Frank & Ernst, 2018; Frank et al., 2016; Vasishth et al., 2010)
- But the perceived difficulty of the structure had an effect; the illusion may indeed be cross-linguistic (Gibson & Thomas, 1999), appearing in the relatively more difficult constructions (e.g., DISC conditions in **Expt. 2**)